2020 AMC 10B Problem 7 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10B Problem 7. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10B Problem 7

Problem:
How many positive even multiples of 3 less than 2020 are perfect squares?

Answer Choices:

A. 7
B. 8
C. 9
D. 10
E. 12

Solution:

Any even multiple of 3 is a multiple of 6, so we need to find multiples of 6 that are perfect squares and less than 2020. Any solution that we want will be in the form $(6n)^2$, where n is a positive integer. The smallest possible value is at $n = 1$, and the largest is at $n = 7$ (where the expression equals 1764). Therefore, there are a total of $(\mathrm{A})\boxed{7}$ possible numbers.

$$\textbf{OR}$$

An even multiple square of 3 can be represented by

$$3^2 \cdot 2^2 \cdot x^2$$

where 3^2 is the multiple or 3 and 2^2 makes it even. Simplifying we have $36^2 \cdot x^2$. We can divide 2020 by 36 (floor) and get 56 . We can then see that there are $(\text{A})\boxed{7}$ different values for x. It can't be larger or else $x^2 > 56$.